Souvik Saha

Date of birth: 04/03/1997 | **Nationality:** Indian | **Phone number:** (+49) 15510509313 (Home) | **Email address:** souvikssaha@gmail.com | **Email address:** st190810@stud.uni-stuttgart.de | **Website:** https://souviksaha97.github.io/ |

Address: Allmandring 3-H, Zimmer - 042, 70569, Stuttgart, Germany (Home)

WORK EXPERIENCE

07/05/2024 Lichtenau, Germany
WERKSTUDENT ATMOS SPACE CARGO GMBH

Supporting the Avionics team by testing and developing code for space applications.

12/2023 – 04/2024
STUDENTISCHE AUSHILFE SLEEPIZ AG

06/2021 – 09/2023 Pune, India
EMBEDDED SOFTWARE DEVELOPER SLEEPIZ INDIA PVT. LTD.

- Responsible for the development and maintenance of embedded software and reduced device recalls from pre-production to deployment by 50%
- Identified requirements for new features adhering to IEC 60601-1, IEC 60601-1-2 and FDA
- Coordinated embedded software verification, release, and deployment activities for a fleet of over 1000 devices in India, Europe and USA
- Established and maintained technical documentation for embedded software in accordance with ISO 13495 and testing as per IEC 62304

07/2019 – 06/2021 Mumbai, India
EMBEDDED SYSTEMS ENGINEER AYU DEVICES PVT. LTD

- Successfully implemented noise cancellation algorithms to reduce background noise by -3dB
- Helped in device development by selecting high quality audio components, reducing noise by -6dB
- Improved the team's efficiency by 30% through automating various manual tasks in the testing process
- Contributed to the development of an automated testing harness reducing test times by 50%

EDUCATION AND TRAINING

07/2015 – 06/2019 Mumbai, India
BACHELOR OF ENGINEERING Vivekanand Education Society's Institute of Technology

- Control Theory
- Advanced Microcontrollers
- Industrial Automation Systems
- Signal Processing
- Circuit Design

Website http://vesit.ves.ac.in/ | **Field of study** Instrumentation Engineering | **Final grade** 7.86/10 |

Thesis Electronic DC Load with Charging/Discharging Modes

10/2023 – CURRENT Stuttgart, Germany
MASTER OF SCIENCE University of Stuttgart

- Embedded Systems Engineering
- Sensor Principles and Integrated Circuits
- Software Engineering for Real-Time Systems
- Advanced Mathematics
- Signal Processing for Human Brain Potentials (EEG)

Field of study INFOTECH | **Final grade** 2.9

LANGUAGE SKILLS

English

Proficiency: Expert (Overall Band 8.5)

DIGITAL SKILLS

C | Python | Linux | Embedded C | ARM | Rust | Embedded Systems | Go | FreeRTOS

PUBLICATIONS

2021
Dubey, S G, S Sudhakar, and S Saha. 2021. "Airflow Noise Reduction in Digital Stethoscope Add-on Device."

IOP Conference Series. Materials Science and Engineering 1070 (1): 012109–9. https://doi.org/10.1088/1757-899x/1070/1/012109.

Open Source Work

Published Arduino libraries to interface IC MCP3202 and IC DAC7611 to Arduino compatible boards.

PROJECTS

06/2018 – 03/2019
100W Electronic DC Load with Charging and Discharging Modes

Worked in a team of 4 to develop an electronic DC Load, which can cyclically charge and discharge a battery at a maximum of 100 W and prepare its discharge curve in real time. Project developed for Final Year Thesis.

Link https://bit.ly/3BumYZh

08/2017 – 02/2018
Texas Instruments India Innovation and Design Challenge 2017

Led a team of 4 and proposed business model for the idea 'Portable Railway Track Flaw Identifier' after market research. Team reached the quarterfinals stage.

08/2016 – 03/2017
eYantra Robotics Challenge 2016

Worked on the Firebird V platform based on Atmega2560 and utilized OpenCV on Python to navigate a robot through a grid to perform pick and drop operations based on object size, shape and colour. Team reached the quarter finals of the competition.

HOBBIES AND INTERESTS

Teaching, Running, Hiking, Squash, Football, Piano, DIY Electronics